

Pranav Kalra

Dehradun, Uttarakhand

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EDUCATION

Vellore Institute of Technology, Vellore

B.Tech in Computer Science and Engineering | CGPA: 9.03 | Top 15% of Cohort

Year of Passing: 2027

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Machine Learning, Statistics

2023 – Present

EXPERIENCE

Accenture

05/2026 – 07/2026

Management Consultant Delivery Associate Intern - Supply Chain & Engineering

Chennai, India

- Analyzed **80+ files and thousands of records** across Item, Location, BOM, Routing, Supplier and Distribution Lane datasets for an **o9 supply-chain implementation**, identifying data gaps and relationship breaks affecting **planning and demand supportability**.
- Decomposed **end-to-end fulfillment paths** into linked data objects, applying a **Profile → Validate → Reconcile → Document** framework to perform **root cause analysis**, isolate failure points and translate technical findings into actionable insights for client stakeholders.
- Identified recurring issues across **master-data integrity, supplier/lot-size logic** and **network completeness**, including a **4,000+ row hierarchy-mapping anomaly**, documenting row-level evidence, impact and ownership to support iterative resolution toward **95%+ demand supportability**.

Evon Technologies

05/2025 – 07/2025

Software Development Engineer Intern

Dehradun, India

- Identified a gap in structured scheduling and event management for a **250+ member tennis community**; scoped, built and shipped **UTTA Tennis Rally Hub**, a production full-stack portal now serving as the group's **primary booking tool**.
- Engineered real-time **event management and scheduling modules**, reducing administrative overhead by an estimated **35%** for organizers.
- Implemented secure **JWT-based authentication** with role-based access control, adhering to **OWASP** security best practices to safeguard data across all user tiers.
- Deployed to **cloud infrastructure via Vercel** with **CI/CD** pipeline integration, achieving **99%+ uptime** and sub-second load times.

Collab Junction

06/2024 – 08/2024

Business Development Intern (Data Analysis & Outreach)

Remote

- Drove data-driven outreach campaigns contributing to **18+ partnership closures**, expanding the collaborator network by **30%**.
- Analyzed lead generation funnels and optimized targeting strategies, improving campaign conversion rate by **28%**.

PROJECTS

CodeSense AI – Automated PR Review Pipeline | Python, FastAPI, Redis, ChromaDB, LLM, RAG | [Link](#)

2026

- Architected an event-driven AI code review pipeline, intercepting **GitHub pull requests** via **webhooks**, queuing jobs in **Redis** and delivering automated inline feedback in **~11 seconds**.
- Engineered a **hybrid RAG engine** combining **Tree-Sitter AST** parsing across **4 languages** with **CodeBERT** embeddings in **ChromaDB**, fusing dense vector similarity with sparse BM25 retrieval via **Reciprocal Rank Fusion** to enrich LLM prompts with codebase context.
- Built an automated **diff-validation guard** that verifies AI comments strictly target modified code, eliminating hallucinated line numbers and preventing **GitHub API errors**.

MedDenoise - Medical Image Denoising System | Deep Learning, Python, PyTorch, FastAPI, Docker, React | [Link](#)

- Trained a lightweight **NAFNet**(2.94M params) on **8000 NIH chest X-rays** using **dynamic AWGN augmentation** (sigma=25), leveraged **SimpleGate** (channel-split element wise multiplication) to eliminate non-linear activation latency, boosting quality from **20.2 dB to 38.47 dB PSNR** while preserving delicate trabecular bone patterns.
- Conducted an **ablation study** comparing **L1 vs. L1+SSIM** loss functions, discovering pure **L1** achieved **superior convergence (38.47 dB vs. 38.37 dB)** by avoiding gradient conflicts on grayscale medical textures.
- Deployed the **PyTorch model** via a **containerized FastAPI** and Docker service, engineering a **16-pixel reflection padding** pipeline to eliminate U-Net dimension mismatch crashes across **arbitrary X-ray resolutions**.

CryptoShield – Real Time Crypto Fraud Detection Pipeline | Python, Apache Spark, Apache Kafka, MongoDB, Docker, Streamlit | [Link](#)

2026

- Built an event-driven fraud surveillance pipeline ingesting concurrent social interaction and OHLCV price streams via Apache Kafka, processing **10-second micro-batches** in **Apache Spark Structured Streaming** to detect coordinated **pump-and-dump manipulation**.
- Designed **graph topology algorithms** using in-memory **adjacency lists** to identify bot rings(**clique density > 60%**) and pump organizers(star topology with **>80% outgoing message ratio**), catching manipulative interaction patterns before market dumps occur.
- Correlated social bot alerts with financial price surges across **38,000+ historical tweets** and **23 cryptocurrency feeds**; stored alert documents in MongoDB and tracked live **Precision/Recall** metrics on a **containerized Streamlit** dashboard.

TECHNICAL SKILLS

Languages: C++, Java, Python, SQL, C, HTML, CSS

Frameworks & Libraries: React.js, REST APIs, Streamlit, FastAPI

AI/ML: Scikit-learn, XGBoost, PyTorch, Pandas, NumPy, SMOTE, LLM Fine-tuning, RAG, Hugging Face

Databases: PostgreSQL, Supabase, SQL (Joins, Aggregations, Indexing), Redis, ChromaDB

Cloud & Tools: AWS (Cloud Practitioner), Vercel, Git, GitHub, Figma, Postman, JWT Authentication, MS Excel, o9

Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks, System Design

Certifications: AWS Certified Cloud Practitioner – Amazon Web Services (2026) | Machine Learning – SmartBridge in collaboration with Google for Developers (2025)

ACHIEVEMENTS

Ideathon Finalist – 3rd Place out of 200+ teams

2024

E-Cell VIT Ideathon | Recognized for technical depth and solution feasibility

VIT Vellore

- Presented a data-driven business solution before a panel of **industry judges and venture capitalists**.

VOLUNTEERING & LEADERSHIP

- Mentorship & Outreach Lead, Collab Junction** (2024): Onboarded **20+ speakers**, designed outreach templates adopted org-wide, and mentored junior interns on lead qualification workflows.
- Core Member, AI/ML Club, VIT Vellore**: Collaborated on club-level ML initiatives, participated in hands-on workshops covering model building and real-world datasets, and contributed to knowledge-sharing sessions among a community of **100+ tech enthusiasts**.